Topic 4 - Retrieval-Augmented Generation for Knowledge-Intensive Tasks

Retrieval-Augmented Generation (RAG) integrates neural retrieval with large language models to address knowledge-intensive NLP tasks. Instead of relying solely on parametric knowledge, RAG systems retrieve external documents to ground generation. This project aims to systematically investigate how retrieval quality, retrieval configuration, and generation strategies jointly influence answer accuracy, faithfulness, and robustness. Using benchmark datasets from different domains, such as Natural Questions (Kwiatkowski et al.), PubMedQA (Jin et al.), and FinanceBench (Islam et al.), you will evaluate how retrieval and generation performance varies across domains with different knowledge structures and terminology. Metrics may include answer accuracy, evidence grounding quality, citation precision, and hallucination rate.

The project topic focuses on two key components:

1. **Retrieval-Generation Interaction:** Investigate different retrieval strategies (sparse/ dense/ hybrid retrieval) paired with various LLMs (e.g., open-source models like LLaMA or closed-source models like GPT). You should:
    - Implement at least 2 retrieval methods (e.g., BM25 vs. Contriever) and 2 generation models
    - Analyze how retrieval precision affects final answer quality using the HotpotQA dev set
    - Compare zero-shot vs. instruction-tuned models (e.g., https://huggingface.co/Intel/neural-chat-7b-v3-3)
    - Conduct experiments on at least two different domains and report cross-domain performance differences.
2. **Hallucination Mitigation:** Design experiments to measure how RAG reduces model fabrication:
    - Quantify hallucination rates using metrics like FActScore (Min et al.)
    - Evaluate whether citation grounding quality differs across domains.
    - Evaluate citation quality through human assessment (e.g., citation precision/recall)
    - Compare vanilla RAG vs. advanced variants (e.g., Self-RAG (Asai et al.))

Students are encouraged to explore innovative retrieval strategies (e.g., query rewriting) or propose new evaluation frameworks.
Hint: Limit retrieved documents to 3–5 per query to balance performance and computational cost.

Reference

1. Lewis P, et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. NeurIPS 2020.
2. Izacard G, et al. Leveraging passage retrieval with generative models for open domain QA. EACL 2021.
3. Asai A, et al. Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection. ICLR 2024.
4. Min S, et al. FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation. EMNLP 2023.